Kosmic Minute: Launch Month Scripts

Oct 17, 2026 · @shae

Our niche

Kosmic Minute: this week's wildest real science, explained in plain English, with the receipts. We take a science story people are already curious about, often one from the news that week, and turn it into a 10-minute story anyone can follow, with every claim linked to its source.

The gap we fill. Big science channels like Kurzgesagt post a few times a year and rarely chase the news. Most faceless channels recycle the same evergreen topics with robot voices, the exact content YouTube cut back in its January 2026 crackdown. Almost nobody explains a new study within a week, in plain words, with sources shown.

What most channels do	What Kosmic Minute does
Recycle the same evergreen topics	Pair every evergreen topic with a fresh news hook ("a study last week found...")
AI or robot voice	Your real voice, a big part of what keeps us safe from YouTube's mass-produced content rules
"Scientists say..." with no proof	Sources linked in every description, and the key study shown on screen
Hype titles the video can't back up	Honest titles: if scientists aren't sure, we say so, and that becomes part of the story
No personality	A dry, grown-up sense of humor: a few well-timed one-liners and one punchline per episode, in your voice
Lectures	Story structure: cold open, cliffhanger before every ad, a myth-buster, a twist ending

Who we're for: curious people about 16 to 35 who love "wait, really?" facts but don't want a lecture.

Rules every episode follows

1. Hook in the first 20 seconds, and a promise of a payoff at the end.
2. Plain words and short sentences. If a 15-year-old wouldn't get it, rewrite it.
3. Every number checked against a source, and every source linked in the description.
4. At least one punchline per episode, dry and smart, never goofy.
5. Two ad breaks, each placed right before a cliffhanger line.
6. The ending line, every time: "Click like, subscribe, and ring the bell so you don't miss a Kosmic Minute."

How to read the scripts: every **NARRATOR** line is yours to record. *[Visual]* notes are for me when I build the video. **AD BREAK** marks where YouTube places a mid-roll ad.

Episode 1 (Oct 25): What If Earth Stopped Spinning for 1 Second?

Thumbnail text: "1 SECOND." over a cracked Earth. **Length:** about 1,350 spoken words, roughly 10.5 minutes.

Script

[COLD OPEN]

NARRATOR: Right now, you're moving. Not walking. Not driving. Sitting perfectly still, you're moving. If you live in the middle of the United States, Earth's spin is carrying you east at around 800 miles an hour. At the equator, it's about 1,000. You don't feel a thing, because everything around you is moving with you. The air. The oceans. Your chair. Your coffee.

NARRATOR: So what happens if all that stops? Not slowly. Just for one second. One tiny pause. By the end of this video, you'll know why that one second would be the worst second in the history of the planet, and why Earth's spin is already changing right now.

[Visual: slow push-in on a realistic Earth at night, city lights. A speed readout ticks in the corner: "800 mph... and you can't feel it." Title card.]

[CHAPTER 1: How fast are we really going?]

NARRATOR: Earth turns once a day. That sounds slow. But Earth is big, about 25,000 miles around at the equator. To cover that in 24 hours, the ground at the equator has to move about 1,000 miles an hour. Faster than a passenger jet.

NARRATOR: The farther you are from the equator, the smaller your circle, so the slower you go. Near the North Pole, you'd barely be turning at all. That's why your speed depends on where you live.

[Visual: globe with latitude rings lighting up. Labels: equator 1,000 mph, Texas about 900, New York about 790, Alaska about 500.]

NARRATOR: So yes, you've been doing 800 miles an hour this whole time. And you still can't find a parking spot.

[CHAPTER 2: The first second]

NARRATOR: Now imagine the solid planet stops instantly. Here's the problem. Only the rock stops. Everything not bolted down keeps going, because of momentum. A moving thing wants to keep moving.

NARRATOR: Think of a bus slamming on its brakes. The bus stops, but the passengers keep moving and get thrown forward. Now picture the whole planet as that bus. People, cars, trees, even buildings would keep racing east at hundreds of miles an hour. Anything not anchored deep in solid rock would be thrown across the ground.

[Visual: slow-motion simulation of a quiet street. A tree, a car, and a mailbox all lurch east together. Keep it dramatic but not graphic.]

[CHAPTER 3: The wind]

NARRATOR: Next comes the air. Our whole atmosphere is spinning along with Earth. If the ground stops, the air doesn't. Near the equator, you'd suddenly have wind blasting across the ground at close to 1,000 miles an hour.

NARRATOR: For comparison, the strongest wind gust ever measured at the surface was 253 miles an hour, during a cyclone in Australia in 1996. That gust ripped instruments apart. Now imagine winds four times faster, everywhere along the middle of the planet, all at once. Friction with the ground would heat the air. Dust and debris would be scoured off the surface.

[Visual: map of Earth with wind arrows blasting east, strongest at the equator, fading toward the poles. Inset: the 253 mph record, labeled.]

AD BREAK. Lead in with: "And the wind isn't even the scariest part. Wait until you see what the oceans do."

[CHAPTER 4: The oceans move]

NARRATOR: Oceans are heavy, and they're moving east just as fast as everything else. Stop the planet, and the water keeps going. It would pile up against the eastern edges of the oceans and roll over coastlines as enormous waves, far bigger than any tsunami in recorded history.

NARRATOR: Places like the west coasts of Europe and Africa, which face east-moving water from the Atlantic, would take the first hit. Low-lying cities near the equator would be in serious trouble within minutes.

[Visual: realistic ocean surface surging forward, then a top-down map showing water piling against coastlines.]

[CHAPTER 5: Myth vs. fact]

NARRATOR: Quick myth check. A lot of people think that if Earth stopped spinning, we'd all float off into space. Not true. Gravity doesn't come from spin. It comes from Earth's mass. Spin actually makes you weigh a tiny bit less, because it slightly flings you outward. At the equator, it's about one third of one percent. So if Earth stopped, you'd actually weigh a hair more. You'd still have a terrible day, but you'd stay on the ground.

NARRATOR: Good news: you'd weigh slightly more. Bad news: so would everything flying toward you.

[CHAPTER 6: What if it stopped slowly?]

NARRATOR: Okay. A sudden stop is impossible. Nothing in physics could brake a planet that fast. So let's try a gentler version. What if Earth slowed down over many years, until it stopped completely?

NARRATOR: Here's something wild. Earth isn't a perfect ball. Because it spins, it bulges at the middle. It's about 27 miles wider across the equator than from pole to pole. The oceans bulge out there too. Take away the spin, and that water flows downhill, toward the poles.

NARRATOR: In 2012, a researcher at the mapping company Esri modeled what happens. The oceans drain away from the middle of the planet and pool into two giant polar oceans. Around the equator, a single enormous strip of dry land appears, wrapping all the way around the world.

[Visual: Earth slowly flattening its bulge. Blue water draining toward north and south, a brown belt of land emerging around the middle.]

AD BREAK. Lead in with: "But the strangest change wouldn't be the map. It would be the sky."

[CHAPTER 7: A year-long day]

NARRATOR: If Earth stopped turning, we'd still go around the Sun. So one side would face the Sun for about six months straight. Then six months of night. One sunrise a year.

NARRATOR: Six months of nonstop sunshine would bake the day side. Six months of darkness would freeze the night side. The weather would change too. Right now, Earth's spin is what makes hurricanes swirl. Without it, winds and storms would behave in ways we've never seen. Where we live and how we grow our food would all have to change.

[Visual: time-lapse of a single sunrise stretched across a whole year. Half the globe glowing hot orange, the other half icy blue.]

NARRATOR: Six months of daylight sounds great. Until you remember the other six months.

[CHAPTER 8: Earth's spin is already changing]

NARRATOR: Now for the real-world part. Earth's spin is not perfectly steady, and scientists measure it with atomic clocks down to fractions of a millisecond.

NARRATOR: Over very long stretches of time, the Moon's pull on our oceans acts like a brake, slowly stretching our days. Hundreds of millions of years ago, days were noticeably shorter than 24 hours.

NARRATOR: But lately, Earth has been spinning a tiny bit faster. On July 10, 2025, we had the shortest day ever measured. It was 1.37 milliseconds shorter than 24 hours. That's far too small to feel, but atomic clocks can catch it.

NARRATOR: Here's the strange part. Melting ice is slowing the spin back down. As polar ice melts, the water spreads out toward the equator. It's like an ice skater spinning, then stretching out her arms. She slows down. A 2024 study in the journal Nature found this effect is big enough to change our clocks. Someday, timekeepers may need to remove one second from the world's clocks. It's called a "negative leap second." Because of the melting ice, that's now expected around 2029 instead of 2026.

[Visual: an atomic clock readout. A graph of day length wiggling up and down. A figure skater spinning slower as she opens her arms.]

[CHAPTER 9: Quick-fire facts]

NARRATOR: Rapid round.

1. Earth also flies around the Sun at about 67,000 miles an hour.

2. Since 1972, timekeepers have added 27 extra leap seconds to keep clocks in line with Earth.
3. In 2022, the world's timekeepers voted to change how leap seconds work starting in the 2030s, making them much less likely.
4. A day on Venus lasts longer than its year.

[Visual: one bold full-screen card per fact, quick cuts, a sound hit on each.]

[FINAL THOUGHT]

NARRATOR: So, if Earth stopped for one second? Thousand-mile-an-hour winds. Oceans flying over continents. Everything not bolted to bedrock thrown east. One second, and the world as we know it would be gone. Which makes it pretty amazing that right now, this very second, it's carrying you at hundreds of miles an hour, and you didn't even spill your drink.

NARRATOR: Well. Most of you didn't.

NARRATOR: This is Kosmic Minute. Click like, subscribe, and ring the bell so you don't miss a Kosmic Minute.

Punchline: "So yes, you've been doing 800 miles an hour this whole time. And you still can't find a parking spot."

YouTube description (paste as is)

> What would happen if Earth stopped spinning, even for one second? Thousand-mile-an-hour winds, runaway oceans, and a day that lasts a year. Plus the real news: Earth's shortest day ever was recorded in 2025, and melting ice is changing our planet's spin.
>
> Sources: Shortest days on record (2025, 2026): https://www.rivieramayanews.mx/earth-records-shortest-day-of-2026-on-july-26/ Melting ice and leap seconds (Agnew, Nature, 2024), via AP: https://ny1.com/nyc/all-boroughs/ap-top-news/2024/03/27/a-faster-spinning-earth-may-cause-timekeepers-to-subtract-a-second-from-world-clocks Earth's shortest day and the Moon's role: https://skyatnightmagazine.com/news/earth-shortest-day-in-history
>
> Click like, subscribe, and ring the bell so you don't miss a Kosmic Minute.
>
> #science #earth #space #physics

Tags: what if earth stopped spinning, earth stops spinning, earth rotation, shortest day ever, leap second, science explained, what if, kosmic minute

Facts checked: equator speed about 1,000 mph. Record gust 253 mph (Barrow Island, Australia, 1996). Equatorial bulge about 27 miles.

day July 10, 2025 (1.37 ms). Negative leap second around 2029 (Agnew, Nature 2024). 27 leap seconds added between 1972 and 2016; 2022 decision to change the leap-second rules from the 2030s. Esri ocean model (Witold Fraczek, 2012).

Episode 2 (Nov 1): Hitler's Nuclear Reactor: How Close Did It Really Get?

Thumbnail text: "HOW CLOSE?" over a glowing metal cube. **Length:** about 1,400 spoken words, roughly 11 minutes.

Note on tone: this is a history-and-science story. Keep the visuals documentary-style: archive photos, maps, diagrams. No swastika imagery in the thumbnail, since that can limit ads.

Script

[COLD OPEN]

NARRATOR: In April 1945, American soldiers entered a cave beneath a castle in a small German town called Haigerloch. Inside, they found something the Allies had feared for years: a nuclear reactor, built by Nazi Germany's top physicists.

NARRATOR: For 80 years, people have argued about one question. How close did Hitler's scientists actually get? The man in charge said they were "just short of success." This month, a team of scientists finally ran the numbers, using the actual uranium cubes from that reactor. The answer is not what he claimed.

[Visual: black-and-white archive photo of the Haigerloch cave entrance. Slow push-in on a single dull metal cube. Title card.]

[CHAPTER 1: The discovery that started it all]

NARRATOR: It began in Berlin, in December 1938. Chemists Otto Hahn and Fritz Strassmann found that when they hit uranium with neutrons, the uranium atoms split apart. Physicists Lise Meitner and Otto Frisch figured out what that meant: each split releases a huge amount of energy, and more neutrons. Those neutrons can split more atoms. That's a chain reaction.

NARRATOR: Scientists everywhere understood the danger at once. Control the chain reaction, and you have a power source. Let it run wild, and you have a bomb. And the discovery had been made in Nazi Germany.

[Visual: animated uranium atom splitting, releasing neutrons that hit more atoms, branching outward.]

[CHAPTER 2: The Uranium Club]

NARRATOR: Germany quickly set up a secret nuclear program, nicknamed the Uranium Club. Its best-known leader was Werner Heisenberg, a Nobel Prize winner and one of the most brilliant physicists alive.

NARRATOR: Their first goal was a reactor: a machine that keeps a chain reaction going steadily. It's a key step, because a running reactor can also make plutonium, one possible path to a bomb. To build one with natural uranium, you need a moderator, something that slows neutrons down so they split atoms more easily. The two best choices were very pure graphite, or heavy water.

NARRATOR: And no, heavy water isn't what's left in your water bottle after a month under the car seat. It's water made with a heavier form of hydrogen, and it's very rare.

[CHAPTER 3: The graphite mistake]

NARRATOR: Here's where things started going wrong for Germany. The graphite they could get had tiny traces of boron in it. Boron is a neutron sponge. It soaks them up and kills the chain reaction. So German scientists decided graphite wouldn't work, and bet almost everything on heavy water.

NARRATOR: American scientists, meanwhile, made extra-pure graphite. On December 2, 1942, under a stadium in Chicago, Enrico Fermi's team switched on Chicago Pile-1, the first reactor to reach a self-sustaining chain reaction. Germany was already falling behind.

[Visual: split screen. Left: a German lab with a question mark over a graphite block. Right: Chicago Pile-1 diagram with a green "CRITICAL" label, December 2, 1942.]

AD BREAK. Lead in with: "Heavy water was Germany's big bet. And the Allies were about to go to war against a factory."

[CHAPTER 4: The war over heavy water]

NARRATOR: Heavy water is extremely rare. Germany's supply came from one place: a hydroelectric plant at Vemork, in occupied Norway. The Allies knew it. In February 1943, Norwegian commandos skied through the winter mountains, slipped into the plant, and

blew up its heavy water equipment. Germany rebuilt. Allied bombers hit the plant later that year. Then, in February 1944, Norwegian resistance fighters sank a ferry carrying Germany's remaining heavy water across a lake.

NARRATOR: Every barrel lost set Germany's reactor back. And as we'll see, that turned out to matter even more than anyone knew.

[Visual: map of Norway with Vemork marked. Snowy mountain footage, a ferry silhouette on a dark lake.]

[CHAPTER 5: The reactor in the cave]

NARRATOR: By early 1945, Allied bombs were falling on Berlin, so Heisenberg's team moved south to Haigerloch. In a cave under the castle, they built their final experiment, called B8.

NARRATOR: Picture a giant chandelier. Six hundred sixty-four uranium cubes, each about two inches wide and five pounds heavy, hung on aircraft cable in long chains. The chains were lowered into a tank of heavy water, surrounded by a wall of graphite. The team switched it on. Neutron counts went up. But the chain reaction never sustained itself. It never went "critical."

[Visual: 3D reconstruction of the B8 reactor: chains of cubes lowered into a tank. A neutron counter rising, then flattening.]

[CHAPTER 6: Captured]

NARRATOR: The Allies had a secret team hunting for exactly this, called the Alsos Mission. In late April 1945, they reached Haigerloch. The German scientists had buried the uranium cubes in a nearby field, hidden the heavy water in barrels, and stashed documents in a latrine. The Americans dug it all up and shipped it out. Heisenberg and other top German physicists were taken to a country house in England called Farm Hall, where hidden microphones recorded everything they said.

NARRATOR: In August 1945, those microphones caught their reaction to the news that America had dropped an atomic bomb on Hiroshima. They were stunned. Some didn't believe it at first.

AD BREAK. Lead in with: "So after the war, Heisenberg said his reactor was just short of success. Eighty years later, someone finally checked."

[CHAPTER 7: The cube in the mail]

NARRATOR: Fast-forward to 2013. A physicist named Timothy Koeth, at the University of Maryland, received a heavy metal cube with a note: "Taken from the reactor that Hitler tried to build." It was one of the real Haigerloch cubes. Hundreds were shipped to the United States after the war. Today, researchers only know where about a dozen of them are. The rest are missing.

NARRATOR: Koeth and his team decided to do something nobody had done properly. Measure the real cubes. Dig through the old records. And calculate exactly what that reactor needed to work.

[Visual: close-up of a real cube with hand-filed notches, the handwritten note beside it.]

[CHAPTER 8: The answer]

NARRATOR: Their study was published on October 16, 2026, in the journal PNAS Nexus. The verdict: B8 needed about twice as much uranium as it had. And more than twice as much heavy water. That's roughly twice everything the Norwegian plant produced during the entire war. Germany simply didn't have it, and with the graphite problem, there was no backup plan.

NARRATOR: Here's the twist. An earlier computer estimate had suggested the design might have worked with about 50% more cubes. The new study, using the real cubes and records, says the gap was much bigger. "Just short of success" was a long way off.

[Visual: bar chart. What B8 had versus what it needed: uranium about 2x, heavy water more than 2x. A label on the gap: "Not close."]

[CHAPTER 9: Myth vs. fact]

NARRATOR: Myth: Nazi Germany almost had the bomb. Fact: a working reactor isn't a bomb. It's only a first step. Germany never reached that first step. America had done it more than two years earlier, and went on to build actual weapons by mid-1945.

NARRATOR: In other words, Hitler's secret super-weapon was a uranium chandelier in a cave. Very ominous. Not very functional.

[CHAPTER 10: Quick-fire facts]

1. Each uranium cube weighed about five pounds, about the same as a bag of flour.
2. The cave at Haigerloch is now a museum.
3. Researchers know of only about 14 surviving cubes, out of almost 1,000 Germany made.
4. Some cubes are thought to have ended up in the Soviet Union, and their fate is unknown.

[FINAL THOUGHT]

NARRATOR: For 80 years, the story was that Hitler's scientists came terrifyingly close. It turns out a handful of Norwegian commandos, some impure graphite, and a shortage of very rare water made sure they never got near. Sometimes history turns on big battles. And sometimes it turns on a few barrels of water at the bottom of a lake.

NARRATOR: This is Kosmic Minute. Click like, subscribe, and ring the bell so you don't miss a Kosmic Minute.

Punchline: "Hitler's secret super-weapon was a uranium chandelier in a cave. Very ominous. Not very functional."

YouTube description (paste as is)

> In 1945, Allied soldiers found Nazi Germany's last nuclear reactor hidden in a cave. Its leader, Werner Heisenberg, said it was "just short of success." A new study using the real uranium cubes says otherwise.
>
> Sources: The new study (PNAS Nexus, Oct 16, 2026), via ScienceDaily: https://www.sciencedaily.com/releases/2026/10/261015073218.htm Science: Nazi Germany had no hope of making an atomic bomb: https://www.science.org/content/article/nazi-germany-had-no-hope-making-atomic-bomb-uranium-cubes-reveal Interesting Engineering: How material shortages crushed the program: https://interestingengineering.com/energy/nazi-germany-failed-nuclear-reactor Physics Today: Tracking the journey of a uranium cube: https://physicstoday.aip.org/features/tracking-the-journey-of-a-uranium-cube ScienceAlert: The missing Nazi uranium cubes: https://www.sciencealert.com/scientists-are-searching-for-600-cubes-of-nazi-uranium-that-have-gone-missing-in-the-us
>
> Click like, subscribe, and ring the bell so you don't miss a Kosmic Minute.
>
> #history #science #nuclearphysics #ww2

Tags: nazi nuclear reactor, heisenberg, haigerloch, uranium cubes, german atomic bomb, ww2 science, manhattan project, kosmic minute

Facts checked: fission discovered December 1938 (Hahn, Strassmann; explained by Meitner, Frisch). Chicago Pile-1 critical December 2, 1942. Vemork sabotage February 1943, ferry sunk February 1944. B8: 664 cubes, about 2 inches and 5 lb each, on aircraft cable in heavy water. Alsos at Haigerloch late April 1945; cubes buried in a field. Koeth received a

cube in 2013. PNAS Nexus study October 16, 2026: about 2x uranium, more than 2x heavy water (about 2x Norsk Hydro's wartime output). Earlier Monte Carlo estimate: about 50% more cubes. About 14 known cubes out of almost 1,000.

Episode 3 (Nov 8): We Punched an Asteroid in 2022. Now We're Going Back to See the Damage.

Thumbnail text: "WE HIT IT." with a spacecraft streaking toward a rocky moon. **Length:** about 1,350 spoken words, roughly 10.5 minutes.

Before posting: check ESA's Hera page for the exact arrival date and any first images. If new pictures are out, I'll swap them into Chapter 8.

Script

[COLD OPEN]

NARRATOR: On September 26, 2022, humanity did something no species on Earth had ever done. We picked a rock in space, flew a spacecraft into it at about 14,000 miles an hour, and changed its path around another asteroid. On purpose.

NARRATOR: It worked even better than planned. But we still don't know exactly what we did to that rock: how big the crater is, what's inside it, or whether we even left a crater at all. This month, a European spacecraft called Hera arrives to find out. And what it learns could one day decide whether a city survives.

[Visual: DART's real final approach images, the small moon growing in the frame until the feed cuts out. Title card.]

[CHAPTER 1: Why we'd ever punch an asteroid]

NARRATOR: Sixty-six million years ago, an asteroid about six miles wide hit Earth near what's now Mexico. It helped wipe out the dinosaurs. Smaller rocks hit far more often. In 2013, a rock about 60 feet wide exploded over Chelyabinsk, Russia. Nobody saw it coming. The shock wave shattered windows across the city and injured about 1,500 people.

NARRATOR: Asteroids are the one natural disaster we could actually prevent, if we spot them early and know how to push them. That second part is the problem. Until 2022, nobody had ever tried.

[Visual: Chelyabinsk dashcam-style recreation of a bright streak across a winter sky. Then a size ladder: 60 ft, 500 ft, 6 miles.]

NARRATOR: The dinosaurs didn't have a space program. We do.

[CHAPTER 2: The perfect target]

NARRATOR: NASA needed a test target that was totally safe. They picked a pair of asteroids. The big one, Didymos, is about half a mile wide. Circling it is a small moon called Dimorphos, about 525 feet across, roughly the size of a football stadium. Neither one is a threat to Earth, before or after the test.

NARRATOR: Here's the clever part. Because Dimorphos orbits Didymos, scientists could measure exactly how long each lap takes. Hit Dimorphos, and if its lap time changes, you've proven you moved it.

[Visual: animation of Didymos with Dimorphos circling it. A stopwatch overlay: "11 hours, 55 minutes per lap."]

[CHAPTER 3: The punch]

NARRATOR: The spacecraft was called DART, the Double Asteroid Redirection Test. It launched in November 2021, about the size of a vending machine with long solar panels. For ten months it flew toward its target. In the last few minutes, it steered itself, with no human help.

NARRATOR: NASA set a goal: change the lap time by at least 73 seconds. DART hit. And the lap time changed by about 32 minutes. From 11 hours and 55 minutes down to about 11 hours and 23 minutes.

[Visual: real DART impact sequence. Stopwatch drops from 11:55 to 11:23, then a stamp: "Goal: 73 seconds. Result: 32 minutes."]

AD BREAK. Lead in with: "So why did it work so much better than anyone expected? The answer is in the debris."

[CHAPTER 4: The rocket-exhaust surprise]

NARRATOR: When DART hit, it blasted a huge spray of rock and dust off the surface. A tiny Italian satellite called LICIACube, which had ridden along with DART, flew by and photographed it. The debris shot out in a giant plume, and a tail of dust later stretched thousands of miles behind the asteroid.

NARRATOR: That flying debris acted like rocket exhaust. When rock flies off one way, the asteroid gets pushed the other way. So the debris gave Dimorphos a much bigger shove than the spacecraft alone. Later, the Hubble Space Telescope even spotted dozens of boulders that had been knocked off and were drifting away.

[Visual: LICIACube's real images of the debris plume. Diagram: arrow of debris one way, bigger arrow pushing the asteroid the other way.]

[CHAPTER 5: Myth vs. fact]

NARRATOR: Myth: we'd just nuke an incoming asteroid, like in the movies. Fact: blowing an asteroid into pieces could turn one big problem into lots of smaller ones, all still headed our way. Scientists much prefer a gentle nudge years in advance. A tiny push early on changes where the asteroid ends up by thousands of miles later. A nuclear option is studied only as a last resort, and even then the idea is usually to explode it near the asteroid, to push it, not to shatter it.

NARRATOR: So the real plan isn't "blow it up." It's "bump it politely, a few years early."

[CHAPTER 6: The questions DART couldn't answer]

NARRATOR: DART was destroyed in the hit, so it couldn't stick around. That left big questions. How heavy is Dimorphos? Without its mass, we can't calculate exactly how much push DART delivered. Is it solid rock, or a loose pile of rubble held together by its own weak gravity? How big is the crater? Or did the impact reshape the whole moon?

NARRATOR: Those answers matter. If a real threat shows up one day, we'll need to know how hard to hit it, and how many times. DART proved the punch works. Hera is here to measure it.

[Visual: Dimorphos as a question mark: half solid rock, half rubble pile.]

AD BREAK. Lead in with: "And Hera isn't going alone. It's bringing two tiny sidekicks."

[CHAPTER 7: Hera and its sidekicks]

NARRATOR: Hera launched from Florida on October 7, 2024, on a SpaceX rocket. On the way, it swung past Mars in March 2025 and grabbed close-up images of Mars' little moon Deimos. It's now arriving at Didymos in November 2026, about a month earlier than first planned.

NARRATOR: Hera carries two small satellites, each about the size of a shoebox. One, called Milani, will map both asteroids and study their dust. The other, Juventas, carries a radar that can look inside Dimorphos, the first radar ever to probe the inside of an asteroid. Then it will try to land on it.

[Visual: Hera render releasing two small cubesats near the twin asteroids. Radar waves passing through Dimorphos.]

[CHAPTER 8: What Hera will measure]

NARRATOR: Over about six months, Hera will map Dimorphos in detail, down to about four inches near the impact site. It will measure the moon's mass, its shape, and what it's made of. It will measure the crater, if there is one.

NARRATOR: Put all that together, and scientists can turn DART from a cool one-time stunt into a real recipe. How big a spacecraft. How fast. How early. That recipe is what planetary defense needs.

[CHAPTER 9: Quick-fire facts]

1. Dimorphos is Greek for "having two forms." Scientists think DART may have actually changed its shape.
2. On April 13, 2029, the asteroid Apophis will pass closer to Earth than many communication satellites. Completely safe.
3. The asteroid 2024 YR4 briefly had about a 3% chance of hitting Earth in 2032. It was ruled out. It still has a small chance of hitting the Moon.
4. NASA is building a telescope called NEO Surveyor to hunt hard-to-see asteroids.

NARRATOR: So scientists named it "two forms." Then they hit it hard enough to give it a second one.

[FINAL THOUGHT]

NARRATOR: For four and a half billion years, Earth has taken whatever space threw at it. In 2022, for the first time, something living on this planet threw something back. Now we're going back to read the results. If a dangerous asteroid ever comes for us, the lessons start here.

NARRATOR: This is Kosmic Minute. Click like, subscribe, and ring the bell so you don't miss a Kosmic Minute.

Punchline: "Scientists named it 'two forms.' Then they hit it hard enough to give it a second one."

YouTube description (paste as is)

> In 2022, NASA's DART spacecraft slammed into an asteroid moon and changed its orbit by 32 minutes, humanity's first test of planetary defense. Now ESA's Hera mission arrives to measure exactly what we did.
>
> Sources: ESA: Hera targets early arrival at Didymos: https://www.esa.int/Space_Safety/Hera/ESA_s_Hera_targets_early_arrival_at_Didymos_asteroids Hera mission overview: https://en.wikipedia.org/wiki/Hera_(space_mission) EarthSky: The Hera mission and the DART impact: https://earthsky.org/space/hera-mission-dart-asteroid-impact/ 2024 YR4 Moon impact odds (Space.com): https://www.space.com/astronomy/asteroids/nasa-raises-the-odds-that-an-asteroid-could-hit-the-moon-in-2032
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> #space #asteroid #NASA #ESA

Tags: DART mission, Hera mission, asteroid impact, planetary defense, Dimorphos, Didymos, NASA asteroid, kosmic minute

Facts checked: DART impact September 26, 2022, about 14,000 mph; orbit 11 h 55 min to about 11 h 23 min (32 min); goal 73 seconds. Dimorphos about 525 ft; Didymos about half a mile. LICIACube imaged the plume. Hera launched October 7, 2024; Mars/Deimos flyby March 12, 2025; arrives November 2026, about a month early; about 6 months of science; cubesats Milani and Juventas; maps to about 10 cm near the impact site. Chelyabinsk 2013, about 1,500 injured. Apophis flyby April 13, 2029. 2024 YR4 peak about 3.1%, later ruled out.

Episode 4 (Nov 15): Voyager 1 Is About to Be One Light-Day Away

Thumbnail text: "1 LIGHT-DAY" beside a tiny spacecraft in deep black space. **Length:** about 1,400 spoken words, roughly 11 minutes.

Timing: post before November 18, the day Voyager 1 crosses one light-day. Searches will spike that week.

Script

[COLD OPEN]

NARRATOR: On November 18, 2026, a machine built in the 1970s will cross a line no human-made object has ever crossed. It will be so far away that a radio message, traveling at the speed of light, takes a full 24 hours to reach it.

NARRATOR: It has less memory than a single photo on your phone. Its radio is about as powerful as a refrigerator light bulb. It was built to last five years. It's been flying for 49. This is Voyager 1, and it's running out of time.

[Visual: real NASA photo of Voyager 1 before launch, then a tiny dot drifting across a vast star field. A counter: "Distance: 1 light-day." Title card.]

[CHAPTER 1: What is a light-day?]

NARRATOR: Light is the fastest thing in the universe, about 186,000 miles every second. Light from the Moon reaches us in just over one second. Sunlight takes about eight minutes. A light-day is how far light travels in a whole day: about 16 billion miles.

NARRATOR: That means when engineers send Voyager 1 a command, they wait a day for it to arrive, and another day for the reply. Every conversation takes two days.

[Visual: a beam of light racing outward, passing the Moon at 1 second, the Sun's distance at 8 minutes, Neptune at about 4 hours, then a long stretch to Voyager at 24 hours.]

[CHAPTER 2: A once-in-176-years chance]

NARRATOR: In the 1960s, NASA engineers spotted something rare. In the late 1970s, the outer planets would line up in a way that happens only about once every 176 years. A spacecraft could swing past one planet and use its gravity like a slingshot to reach the next.

NARRATOR: NASA built two. Voyager 2 launched first, on August 20, 1977. Voyager 1 launched on September 5, on a faster path. That's why it took the lead, and has kept it ever since.

[Visual: animated planet alignment with slingshot paths arcing from Jupiter to Saturn and beyond.]

NARRATOR: Voyager's computers have about 69 kilobytes of memory, millions of times less than your phone. And they still work from billions of miles away. Most of us can't get our printer to work from across the room.

[CHAPTER 3: Rewriting the textbooks]

NARRATOR: In 1979, Voyager 1 reached Jupiter. It found faint rings around the planet. And on Jupiter's moon Io, a navigation engineer named Linda Morabito spotted a strange plume rising off its edge. It was a volcano, erupting. The first active volcano ever seen anywhere beyond Earth.

NARRATOR: In 1980, Voyager 1 reached Saturn and flew close to its giant moon Titan. That close pass bent its path up and out of the plane of the planets. It could never visit another planet. Its next stop would be the space between the stars.

[Visual: Voyager's real images of Jupiter, then Io's volcanic plume. Diagram of the path bending up out of the planets' plane.]

AD BREAK. Lead in with: "On its way out, Voyager turned around for one last look at home. That picture changed how we see ourselves."

[CHAPTER 4: The Pale Blue Dot]

NARRATOR: On February 14, 1990, at the urging of astronomer Carl Sagan, Voyager 1 turned its camera around and photographed the planets from almost four billion miles away. In one frame, Earth is a tiny speck, less than a single pixel, sitting in a beam of scattered sunlight.

NARRATOR: Every person who ever lived, every war, every love story, every empire, all on that speck. Shortly after, the cameras were switched off to save power. They've never been turned back on.

[Visual: the real Pale Blue Dot image. Slow zoom into the speck. Silence for two seconds.]

[CHAPTER 5: Leaving the Sun's bubble]

NARRATOR: The Sun constantly blows out a wind of charged particles, creating a giant bubble around the solar system. Where that wind finally runs into the gas between stars, there's a boundary. In August 2012, Voyager 1 crossed it. For the first time, a human-made object was in interstellar space. Voyager 2 followed in 2018.

[Visual: diagram of the Sun's bubble, Voyager 1 crossing the edge, labeled "August 2012."]

[CHAPTER 6: Myth vs. fact]

NARRATOR: Myth: Voyager 1 has left the solar system. Fact: it depends where you draw the line. It's left the Sun's bubble, so it's in interstellar space. But the Sun's gravity still reaches much farther, out to a vast cloud of icy objects called the Oort Cloud. At Voyager's speed, it won't get through that for tens of thousands of years.

[CHAPTER 7: Fixing a computer from 15 billion miles away]

NARRATOR: In late 2023, Voyager 1 suddenly started sending back nonsense. A tiny piece of one computer's memory had failed. Nobody could fly out to fix it. So engineers, some of them working from decades-old paper documents, figured out how to move the damaged code into other parts of memory. By 2024, Voyager was sending real science again.

NARRATOR: Think about that. A software update, beamed across 15 billion miles, to a computer older than most of the people fixing it. And it worked.

[Visual: real NASA team footage. A stream of garbled data slowly turning into clean numbers.]

AD BREAK. Lead in with: "But there's one problem engineers can't fix. Voyager is running out of power."

[CHAPTER 8: Going quiet]

NARRATOR: Out past Jupiter, sunlight is too weak for solar panels. So Voyager runs on heat from decaying plutonium, turned into electricity. At launch it made about 470 watts. It loses about 4 watts every year.

NARRATOR: To keep it alive, engineers have been switching things off, one by one. Heaters. Backup systems. Science instruments. Voyager 1 launched with ten science instruments. As of 2026, only two are still on. Sometime in the next few years, there won't be enough power to talk to us at all.

[Visual: a control panel with lights going dark one at a time. Final two still glowing.]

[CHAPTER 9: Quick-fire facts]

1. Voyager 1 is moving at about 38,000 miles an hour compared to the Sun.
2. It carries a Golden Record with greetings in 55 languages and about 90 minutes of music, from Bach to Chuck Berry.
3. NASA talks to it through giant antennas in California, Spain, and Australia.
4. In about 40,000 years, it will pass relatively close to another star.

[FINAL THOUGHT]

NARRATOR: Voyager 1 was built to last five years. It's lasted 49. It found volcanoes on another world, took the most humbling photo ever made, and became the first thing we ever sent into interstellar space. Soon it will go silent. But it won't stop. It'll drift through the Milky Way for billions of years, carrying our music, long after everyone who built it is gone.

NARRATOR: And on November 18, it'll be one light-day from home. Not bad for a machine with less memory than one photo on your phone.

NARRATOR: This is Kosmic Minute. Click like, subscribe, and ring the bell so you don't miss a Kosmic Minute.

Punchline: "Most of us can't get our printer to work from across the room."

YouTube description (paste as is)

> On November 18, 2026, Voyager 1 becomes the first human-made object one light-day from Earth: a message takes 24 hours to reach it. Here's how a 1970s spacecraft got this far, how engineers keep it alive, and why it's about to go quiet.
>
> Sources: EarthSky: Voyager 1 one light-day from Earth: https://earthsky.org/space/voyager-1-1-light-day-from-earth-november-17-18-2026/ Scientific American: Voyager 1 down to two science instruments: https://www.scientificamerican.com/article/nasas-voyager-1-spacecraft-down-to-just-two-working-science-instruments/ Discover: As Voyager 1 nears a historic distance: https://www.discovermagazine.com/as-voyager-1-nears-a-historic-distance-nasa-continues-shutting-down-instruments-49387 The Planetary Society: 2026 space events calendar: https://www.planetary.org/articles/calendar-of-space-events-2026
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> #space #NASA #voyager #astronomy

Tags: Voyager 1, light day, Voyager 1 2026, NASA Voyager, interstellar space, pale blue dot, golden record, kosmic minute

Facts checked: one light-day on November 18, 2026; a light-day is about 16 billion miles. Launches: Voyager 2 August 20, 1977; Voyager 1 September 5, 1977. Jupiter 1979 (Io volcano spotted by Linda Morabito); Saturn and Titan 1980. Pale Blue Dot February 14, 1990. Heliopause August 2012; Voyager 2 in 2018. Memory fault late 2023, fixed 2024. RTG output about 470 W at launch, losing about 4 W a year. Two of ten instruments still on as of 2026. Golden Record: 55 languages, about 90 minutes of music. Deep Space Network: California, Spain, Australia.

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